

MARIA PAUL THURKDAYIL

DATA SCIENCE STUDENT



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SUMMARY

Master's student gaining proficiency & building skills in the field of Data Science Prior background in Computer Science and Engineering. Looking for Data Science and Machine Learning Based Roles.

For details on the projects, I worked on, check out github.com/intagliated

EDUCATION

Chennai Mathematical Institute, Masters in Data Science

Aug 2024- May 2026

EXPIERENCE

AI Research Intern , IIT Jammu | June 2026- Present

<https://iitjammu.ac.in/mira/>

1. Synthia - Synthetic Data Generation Pipeline

Worked on a Python pipeline for generating instruction fine-tuning data (context, instruction, response triples) from raw text documents using self-hosted LLMs

Set up and validated the local development environment, connecting to MIRA's private GPU inference cluster over Tailscale VPN.

Mentor : Parth Iyer, CMI

2. Gaperon - Multilingual Text Quality Classifier

Worked on a project using jhu-clsp/mmbert-base with 6 independent classification heads to predict text quality (clarity, coherence, depth, grammar, usefulness, overall) across English, Hindi, and Tamil

Set up end-to-end pipeline using SkyPilot with HuggingFace Hub integration for dataset streaming, checkpoint versioning, and model promotion across branches

Mentor: Samarth Ramesh, CMI

AI Research Intern, S3 Labs, IIT Bhilai | May 2025- July 2025

1. Technical Image Understanding

Evaluating how Packet Diagrams, Sequence Diagrams, State Diagrams, etc are inferred using LLM's through Mermaid Code Diagrams and hand-drawn images

2. Legal Infographic Generation

LLM-powered application to automate infographic generation for legal documents.

Conducted Research Paper Analysis on the Applicability of LLM's in the Legal Domain.

Experimented with Multiple LLM's and developed trial prototypes for generating infographics such as time-lines, key arguments, etc for the legal domain.

Technology: LangChain, GenAI, Hugging Face

Dataset: IL-TUR (Exploration Lab, IIT Kanpur)

Supervisor: Gagan Raj Gupta, Department of Computer Science and Engineering.

PROJECTS

Differential Privacy in PyTorch (Opacus)

Open Source Contribution, Data Privacy and Data Quality Project

May 2026

- Attempted a Contribution to Opacus, Meta's open-source library enabling differentially private deep learning with minimal code changes.
- Worked on an issue that had a privacy bug (silent epsilon miscalculations with WeightedRandomSampler).

Perspective Aware Sentiment Analysis

Applied Data Analytics Project, Sentiment Analysis

November 2025

Models: Roberta-base, GPT-4o-mini

Technology: transformers, openai, feedparser, pandas, python-dotenv

- Real-time data pipeline using Google News RSS to fetch and parse headlines based on specific search queries and time windows.
- Implemented a standard sentiment baseline using the cardiff nlp/twitter-roberta-base-sentiment-latest model to provide objective scores. Perspective-aware sentiment engine was created using GPT-4o-mini that adapts its analysis based on user-defined personas .

Predicting Crime Categories

Ensembled Learning, Law & Order

June 2024- Aug 2024

Models: XGBoost, LightGBM

Technology: xgboost, lightgbm, pandas, scikit-learn

- Dataset consisted of **20,000 rows** & 20 columns about location, victim demographics, date, time, etc. & 6 crime categories for the Target Variable.
- **LGBMClassifier()** and **XGBoostClassifier()** were used in building a **VotingClassifier()** which had an accuracy of **0.88080** in contrast to the baseline accuracy of around **0.58**.

Forecasting the Total Marine Fish Production in Kerala

Development & Comparative Analysis of Machine Learning Models

Jan 2022- May 2022

Machine Learning, Fisheries

Models: Multiple Linear Regression, Multilayer Perceptron, Radial Basis Function, Random Forest

Technology: SPSS, Python- scikit-learn, pandas

- Data on Marine Fish production in Kerala (in tonnes) was collected from **CMFRI, Kochi** and data on 9 significant environmental variables from **ICOADS** from **1960 - 2014** .
- Highest predictive power was for **Random Forest- R2 score 0.87, MAE 0.05, MSE 0.005, RMSE 0.07**

Chat Application- Publisher Subscriber Architecture

November 2019

- Developed a real-time communication system using the MQTT protocol, applying a publisher-subscriber model to support chat-style applications as part of Hack-Kochi#2

TOOLS & LIBRARIES

Data Science Libraries: TensorFlow, OpenCV, scikit-learn, NumPy, pandas, matplotlib, seaborn

Statistical Tools: R, SPSS, statsmodels

Data Analytics: Python, Power BI, MS-Excel

RDBMS: MySQL, PostgreSQL, MongoDB, Redis

Big Data Frameworks: Apache Spark, Hadoop

Model Deployment Frameworks: Flask, Streamlit , Skypilot, Docker , Tailscale

TEACHING & MENTORING

Teaching Assistant, NPTEL

Jan 2024- April 2024

noc24-cs47: The Joy of Computing Using Python.

Instructor: Dr Sudarshan Iyengar, Head of Computer Science Department, IIT Ropar

Prepared weekly assignments & handled the forum for the course having **60,000** registered learners.

Student Mentor, IITM BS Degree Programme

Mar 2024- April 2024

BSMA1004: Statistics for Data Science 2

Instructor: Andrew Thangaraj, Professor, Electrical Engineering, IIT Madras

Conducted doubt clearing sessions of 2 hours, 3 times a week for a set of **25 students**.

OTHERS

1. Winner Hack Kochi #2

November 2019

24-hour Hackathon for girls organised by IEEE Kochi Hub.

2. KEAM Rank 333

August 2018

Kerala State Engineering Entrance Exam.

VOLUNTEERING

Covid-19 Response Volunteer, Caregiver Saathi

May 2021

ONLINE CERTIFICATIONS

- Google IT Support Specialization, Coursera
- Introduction to Linux, The Linux Foundation
- Machine Learning Foundations: A Case Study Approach
- SQL For Data Science